



**ENT-164** · Intro to Making



Derby Entrepreneurship  
Center at Tufts

CLASS 5 · WEEK 5 · FALL 2026

# Electronics

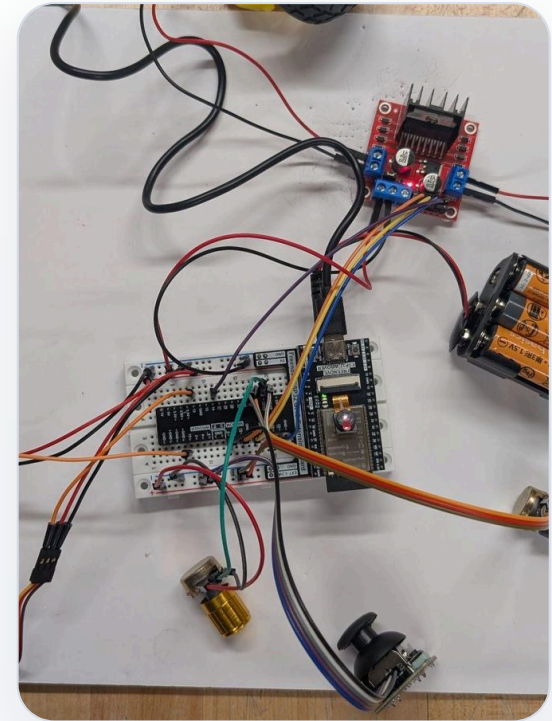
Today your robot gets a brain. We wire motors, sensors, and controls on the breadboard, install MicroPython, and make AI our coding partner.

Thu, Oct 8

JCC 301 / Nolop

Robo Challenge

Bring your ESP32 kit



# Seven weeks from sketch to race day

Today we're at WK4 — electronics. Here's how the robot build comes together.

WK1

## Sketch & Prototype

Sketch concept variations; build a foam core mockup.

WK2

## Laser Cutting

Create a laser-ready 2D design; cut structural parts.

WK3

## 3D Printing

Model 3D parts; 3D print functional parts.

WK4

## Electronics

Wire motors, sensors, and controls; program with MicroPython + AI.

WK5

## Connectivity

Connect to a wireless network; build a remote control interface.

WK6

## Final Testing

Final integration and test runs.

WK7

## Race!

Demonstrate the robot with mechanical systems and motors mounted.

TODAY

**WK4 · Electronics** — wire motors, sensors, and controls on a breadboard, and program basic movements using MicroPython and AI.

# Today the robot gets a brain

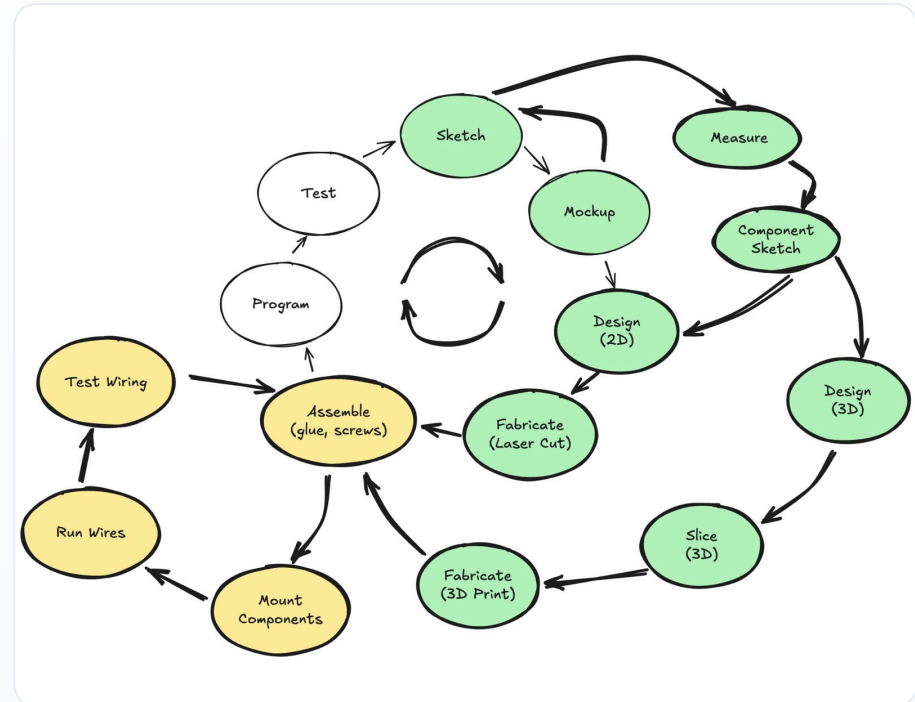
The mechanics are nearly done — today we bring the electronics to life.

## TODAY'S PLAN

- Meet your kit and the ESP32 board
- Install the MicroPython operating system
- Blink an LED from Thonny
- Move a servo with AI-generated code
- Wire the robot at Nolop

## WHERE WE ARE IN THE BUILD

Sketch, mockup, design, and fabrication are behind you. Now we assemble, run wires, test the wiring, program, and test again.



# What's in your **personal kit**?

Freenove FNK0046 Super Starter Kit for ESP32 WROVER — everything you need for the electronics project.

## The brain

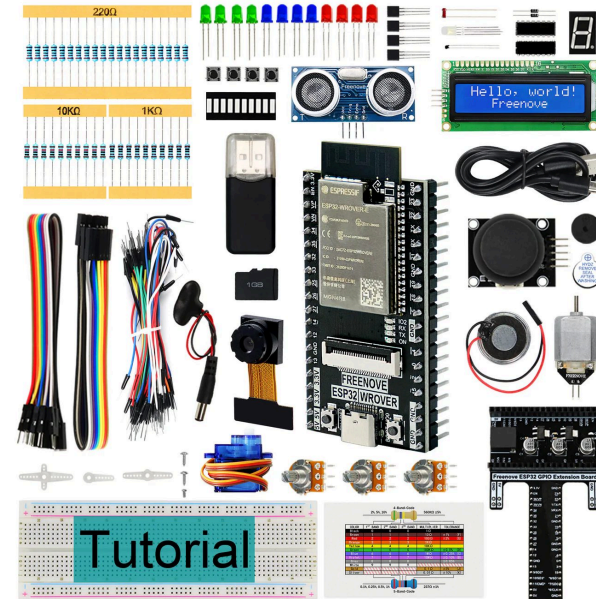
ESP32-WROVER board, GPIO extension board, and a solderless breadboard.

## The basics

Jumper wires, LEDs, resistors (220 $\Omega$ , 1K, 10K), push buttons, USB cable.

## The fun stuff

LCD and 7-segment displays, buzzers, joystick, potentiometers, servo, TT motor + wheel, motor driver, ultrasonic sensor, camera.



### NOTE

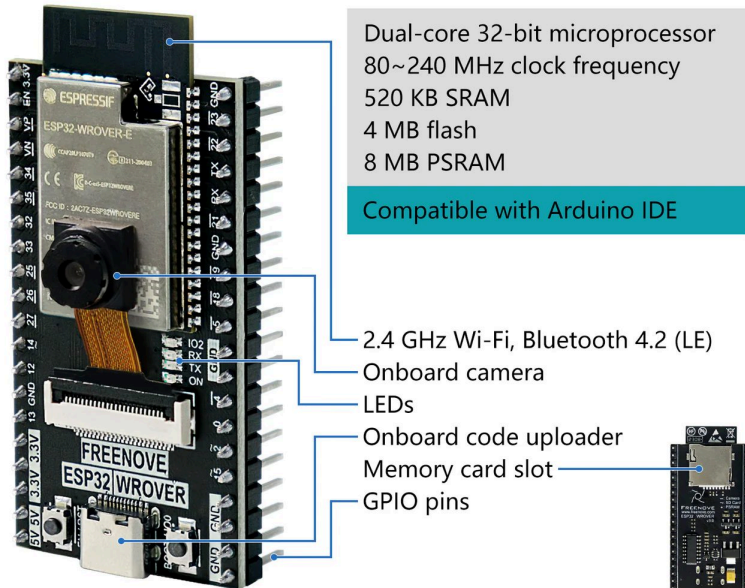
No paper tutorial — the download link for the tutorial and code is **on the box**.

# Meet the ESP32-WROVER

A powerful, small controller with onboard wireless and camera.

## ESP32-WROVER Board

A powerful, generic and small-sized controller with onboard wireless and camera.



### SPECS

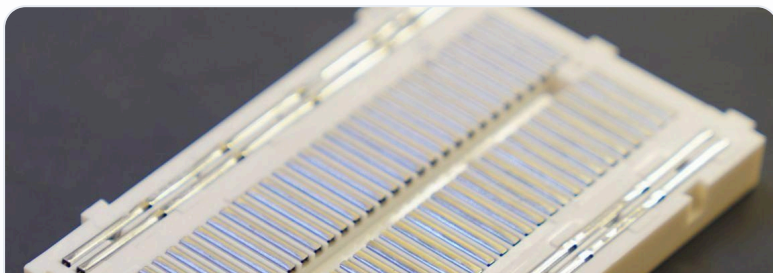
- Dual-core 32-bit microprocessor, 80–240 MHz
- 520 KB SRAM · 8 MB PSRAM · 4 MB flash
- 2.4 GHz Wi-Fi and Bluetooth 4.2 LE
- Onboard camera, LEDs, and code uploader
- Memory card slot and GPIO pins

### WHAT IT CONTROLS

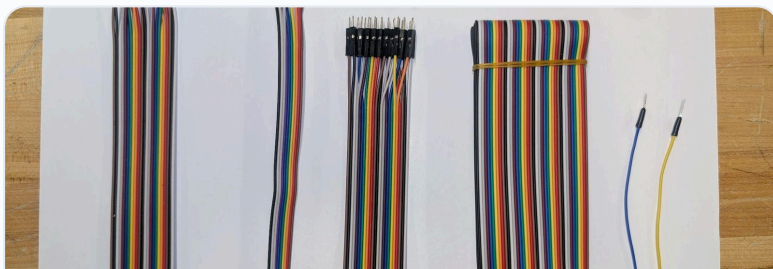
Motors through the driver, servos, joystick, potentiometers, sensors, and displays — the brain of your robot.

# How the **breadboard** works

No soldering — push parts and wires into the holes and they connect.



A solderless breadboard — rows of connected holes



Jumper wires — male–male, male–female, female–female

## Rows

Each row of five holes is connected horizontally, and the center channel separates the two sides.

## Power rails

The vertical rails along the edges carry 5V, 3.3V, and GND to every row.

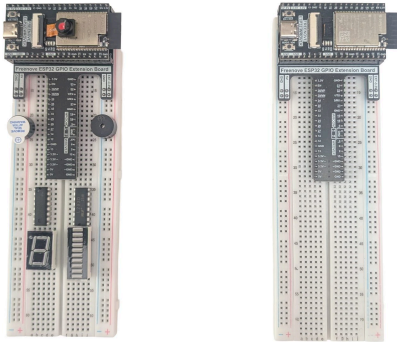
## Pin rows

Your ESP32 plugs into the GPIO extension board: any of the horizontal holes in a pin's row connects to that pin — for example, pin 34.



# Prep your board

Three steps before the first circuit.



1 · Remove the pre-installed demo parts

02

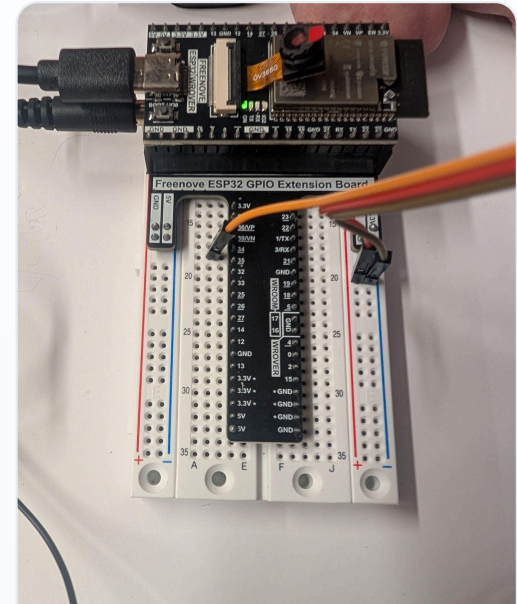
## Seat the board

Clip the ESP32 and its GPIO extension board across the breadboard's center channel.

03

## Read the labels

5V, 3V3, GND, and the numbered GPIO pins are printed on the extension board. Check before you plug anything in.



Pin labels on the extension board

# Installing the **operating system**

Your ESP32 runs MicroPython — a tiny version of Python.

## FOLLOW THE INSTRUCTIONS

"Put the snake on the plane" — flash MicroPython onto your ESP32, then connect with Thonny.

[go.tufts.edu/esp32](https://go.tufts.edu/esp32)

## One-time setup

Do this once per board. After that, Thonny connects and runs your code on the robot.



The MicroPython snake needs a plane — your ESP32 board.



# Make an LED blink

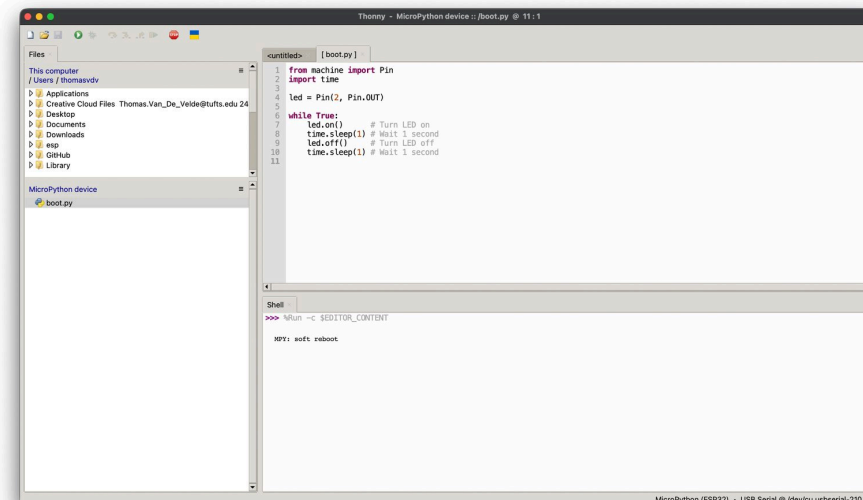
Type it in Thonny (or ask AI for it), then run it on the board.

## BLINK.PY

```
from machine import Pin
import time

led = Pin(2, Pin.OUT)

while True:
    led.on()      # Turn LED on
    time.sleep(1) # Wait 1 second
    led.off()     # Turn LED off
    time.sleep(1) # Wait 1 second
```



## CHALLENGE

Can you make it blink faster or slower?

# Generate code and wiring with AI

Go to [claude.ai](https://claude.ai) and enter this prompt.

## THE PROMPT

I have an ESP32 WROVER with MicroPython and an SG90 servo.

I want to move the servo arm from 0 to 180 degrees and back.

Generate the code and provide beginner-friendly wiring instructions.

✶ Hey there, Maker

I have an ESP32 WROVER with MicroPython and an SG90 servo.

I want to move the servo arm from 0 to 180 degrees and back.

Generate the code and provide beginner-friendly wiring instructions.

+

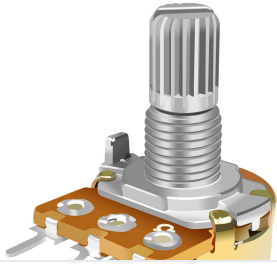
Opus 4.6 ▾



Describe your parts, your goal, and ask for beginner-friendly wiring — one sentence per idea.

# Knobs, sticks, and sensors

Your robot reads the world through inputs — here's how two common ones wire up.



Potentiometer — a knob the ESP32 can read

## POTENTIOMETER

- Left pin → GND
- Middle pin (wiper) → ESP32
- Right pin → 3V3



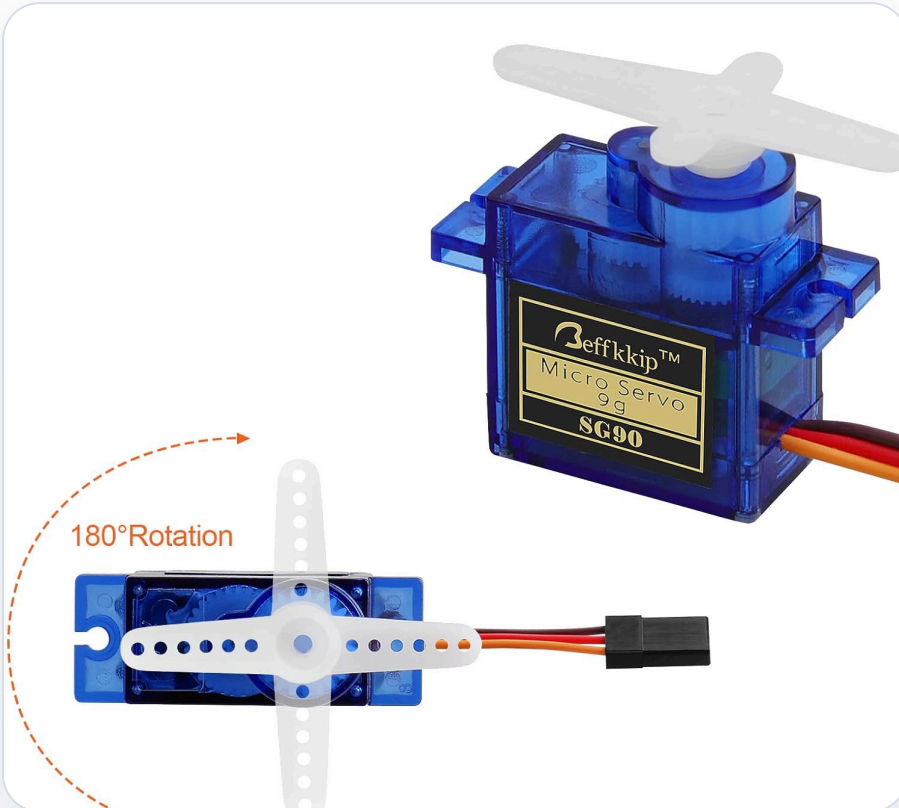
KY-023 joystick — two axes plus a button

## KY-023 JOYSTICK

- GND → GND
- +5V → 3V3
- VRx (X-axis) → ESP32
- VRy (Y-axis) → ESP32
- SW (button) → ESP32

# Move an arm with a servo

The SG90 micro servo rotates about 180° — perfect for arms, gates, and grippers.



## SERVO WIRING

- Brown / Black → GND
- Red → 5V
- Orange / Yellow → ESP32

## ON THE ROBOT

Servo 1 and Servo 2 wire up the same way — match the colors, not the memory.

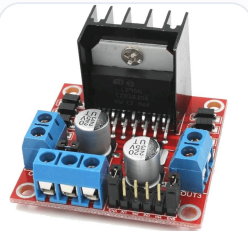
# Spin the wheels with the L298N

The driver is the muscle: the ESP32 sends signals, the L298N sends power to the motors.



## TT Motor

DC gearbox  
motor + wheel



## L298N driver

Dual H-bridge  
motor driver

## L298N WIRING

- 12V → battery pack
- 5V → logic power
- GND → common ground
- IN1, IN2 → ESP32 (left motor)
- IN3, IN4 → ESP32 (right motor)
- OUT1, OUT2 → left motor (red, black)
- OUT3, OUT4 → right motor (red, black)

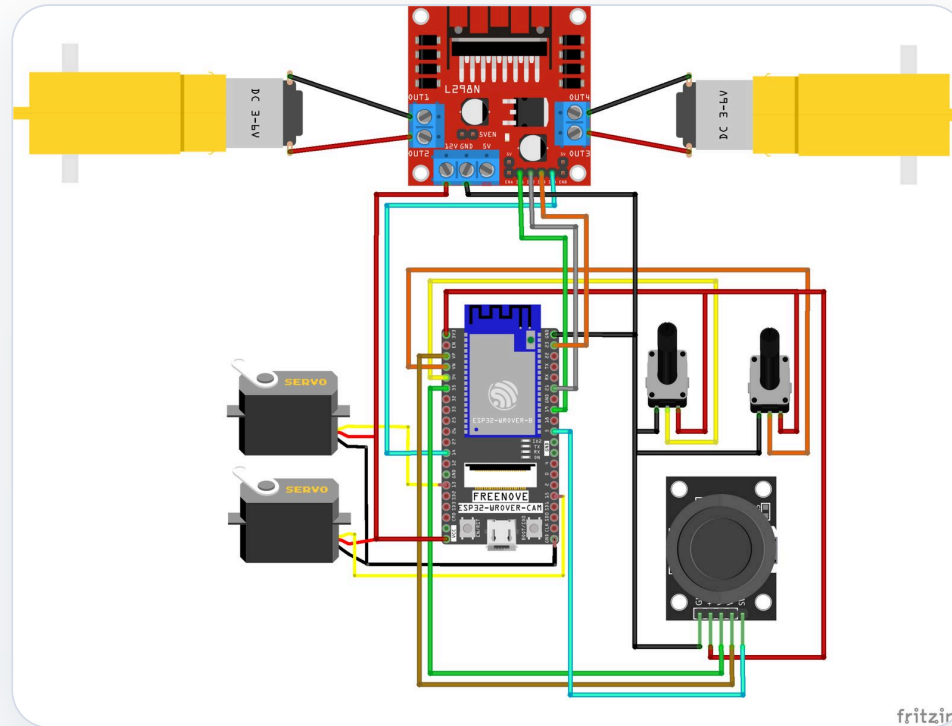
# Your robot's wiring map

Two servos, two DC motors, two joysticks, one driver, one brain.

## READ IT LIKE A MAP

- Power comes from the battery pack
- Motors run through the L298N driver
- Servos and controls go to GPIO pins

Wire from a diagram, never from memory — and ask AI to explain any wire you don't recognize.





# Wire your robot **one part at a time**

Mechanical assembly review, then AI-assisted wiring at the makerspace.

## TODAY AT NOLOP

- Mechanical assembly review
- Use AI to wire robot components one by one
- Mount parts and run wires after functional testing

## TIPS

- Do not generate all the code and wiring in one prompt.
- Add components step by step. Test and verify with each iteration.
- Provide detailed model numbers for the parts.

## NAME YOUR PARTS EXACTLY

- DC Gearbox Motor – "TT Motor"
- BOJACK L298N Motor DC Dual H-Bridge Motor Driver
- KY-023 Control Stick
- SG90 9g Micro Servo Motor
- Freenove ESP32 WROOM

# AI-assisted electronics project

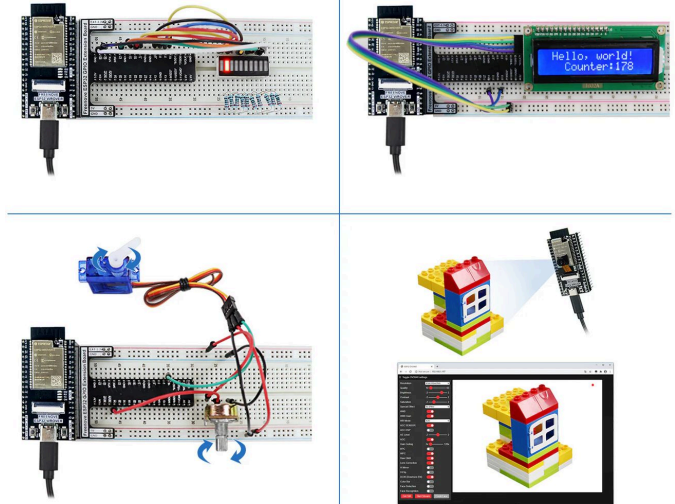
Design and build a fun electronics project using your ESP32 kit — with AI as your coding partner.

## REQUIREMENTS

- At least 1 input (sensor, button, etc.) + 1 output (LED, display, buzzer, etc.)
- AI-generated MicroPython code, tested and refined

## THE PROCESS — ALL IN ONE CHAT

- **Brainstorm:** attach the parts list image, ask AI for project ideas
- **Build:** get wiring instructions and code from AI
- **Test & iterate:** describe problems, share errors, refine with AI's help



Need ideas? The kit tutorial has example projects and code to borrow from.

# What you'll hand in

01

## Your AI conversation

A link to your chat — one continuous conversation, start to finish.

02

## Video demo

A demo of the working project with a voice-over explanation.

03

## Written reflection

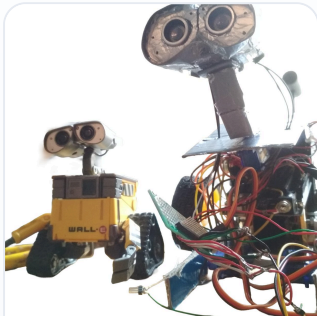
A written reflection on the process.

**DEMO** Bring your electronics project to class for a demo.

**GRADING** You're graded on **collaboration and learning**, not coding skills. Keep one chat, iterate often, and ask lots of questions!

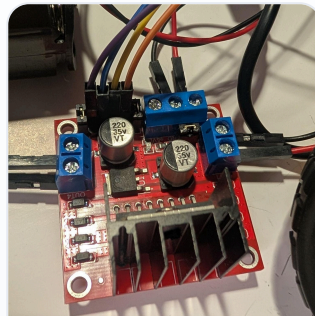
# Robots under construction

Wiring, testing, and mounting — the messy middle of the build.



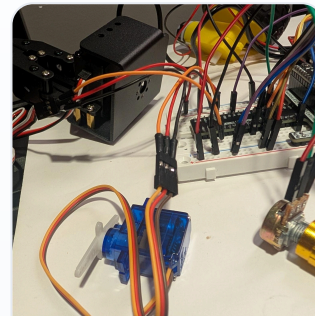
## Wire by wire

The robot takes shape.



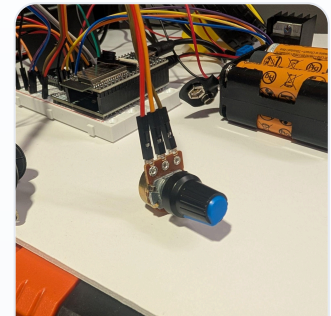
## Motor driver

Powered and wired in place.



## Test rig

Test each part before mounting.



## Controls

Knobs wired and verified.

BY THE END OF CLASS

# Personal reflection

Before you leave, complete your personal reflection for today's class.

[go.tufts.edu/reflect](https://go.tufts.edu/reflect)

BEFORE YOU GO

# Questions?

Next week: Connectivity — connect your microcontroller to a wireless network and build a remote control interface.

## WHERE TO FIND US

### Office hours

[go.tufts.edu/meet\\_tvdv](https://go.tufts.edu/meet_tvdv)

### Email

[thomas.van\\_de\\_velde@tufts.edu](mailto:thomas.van_de_velde@tufts.edu)

### Course materials

[tuftsmaker.github.io/ENT-164](https://tuftsmaker.github.io/ENT-164)